The Amazon SageMaker family, untangled

What each piece is, when to reach for it, and where it sits in the ML lifecycle. Status checked against AWS docs, October 26, 2026.

Closed to new customers · Jul 30 2026 · Moved / renamed · New since 2025 · Heads-up

Overlap group · Same letter = same job, different tool or user. Compare them on the "Same job, different tool" board.

DS data scientist · MLE ML engineer · DE data engineer · Analyst business analyst · Ops labeling & review · Risk governance · Admin platform admin

THREE NAMES PEOPLE MIX UP

Amazon SageMaker
The umbrella platform for data, analytics, and AI (the "next generation" of SageMaker, announced Dec 2024). Not one tool: the whole family on this page.

SageMaker AI
The original build, train, and deploy service, renamed. Runs standalone in its own console or inside Unified Studio. Everything in the seven columns below.

SageMaker Unified Studio
One web workspace that pulls in EMR, Glue, Athena, Redshift, MWAA, Bedrock, and SageMaker AI. Generally available March 2025.

Unified Studio
docs ↗
DE · DS · Analyst · A Workspace · B Data prep
What: Projects, notebooks, SQL editor, visual ETL, and ML tools in one governed workspace.
When: Data, analytics, and ML teams need to share data and work together.

Lakehouse
docs ↗
DE
What: Unified, Apache Iceberg-compatible access to data in S3 and Redshift.
When: Query and train on one copy of data from any engine.

Catalog
docs ↗
DE · Risk · I Track & govern
What: Discovery, sharing, and governance of data and AI assets. Built on DataZone.
When: Control who can find and use which datasets and models.

Data Agent
docs ↗
DE · DS · New · re:Invent 2025
What: AI agent built into Unified Studio notebooks that writes and fixes code.
When: Speed up data prep and analysis inside notebooks.

Bedrock in Unified Studio
docs ↗
Developer · DS · Separate service · D Customize LLM · H Use an FM
What: Foundation models by API, plus agents, knowledge bases, and guardrails.
When: Build GenAI apps without training or hosting models yourself.

SAGEMAKER AI, BY LIFECYCLE STAGE

01 Where you work

Studio
docs ↗
DS · MLE · A Workspace
What: Web IDE hub: JupyterLab, Code Editor (VS Code), RStudio, and every SageMaker AI tool.
When: Default home for code-first ML work.

Notebook Instances
docs ↗
DS · A Workspace
AL2 retired · AL2023 default
What: A single managed Jupyter VM, no Studio domain required.
When: Quick standalone notebooks or older workloads.

Canvas
docs ↗
Analyst · B Data prep · C Build a model
What: No-code visual ML: prep data, build models, predict. Now hosts Data Wrangler and the Autopilot UI.
When: Analysts need predictions without writing code.

HyperPod
docs ↗
MLE · Research · D Customize LLM · E Train at scale
Checkpointless + elastic, 2025
What: Persistent, self-healing accelerator clusters run with Slurm or EKS.
When: Weeks-long distributed training or fine-tuning of foundation models.

Partner AI Apps
docs ↗
DS · MLE
What: Certified third-party ML and GenAI apps that run inside your SageMaker AI domain.
When: You want a partner tool without data leaving your account.

Studio Lab
docs ↗
Learner · A Workspace · Closed to new
What: Free JupyterLab with CPU/GPU time, no AWS account needed.
When: Learning and experimenting at zero cost.

02 Prepare & label data

Data Wrangler
docs ↗
Analyst · DS · B Data prep
Now inside Canvas
What: Visual data prep and feature engineering with built-in transforms.
When: Clean and featurize data with little or no code.

Processing Jobs
docs ↗
DS · DE · B Data prep
What: Run your own prep or evaluation script (scikit-learn, Spark) on managed compute.
When: Repeatable, scripted data prep or model evaluation.

Feature Store
docs ↗
DS · MLE
What: Shared feature repository: online store for low latency, offline store for training.
When: Reuse features across teams; keep training and inference consistent.

Ground Truth
docs ↗
Ops · DS · G Human review · Closed to new
What: Data labeling with human workforces plus automated labeling.
When: You need labeled training data or human feedback.

Geospatial
docs ↗
DS · Closed to new
What: Built-in satellite imagery, geospatial models, and map notebooks.
When: Earth observation, land use, or location-based predictions.

03 Build or pick a model

JumpStart
docs ↗
DS · MLE · D Customize LLM · H Use an FM
What: Hub of foundation models, pretrained models, and solution templates.
When: Start from an existing model; deploy or fine-tune in a few clicks.

Serverless Model Customization
info ↗
MLE · DS · D Customize LLM
New · re:Invent 2025
What: Guided fine-tuning (SFT, DPO, RLVR, RLAIF) of Nova and open-weight models, no infrastructure.
When: Customize an LLM without managing training clusters.

Autopilot
docs ↗
DS · Analyst · C Build a model
UI moved into Canvas
What: AutoML: profiles data, tries algorithms, ranks candidate models.
When: Fast baseline models, via Canvas or the AutoML API.

Built-in Algorithms & Containers
docs ↗
DS · C Build a model
What: Managed algorithms (XGBoost and others) and prebuilt framework containers.
When: Train common model types without building your own container.

04 Train & tune

Training Jobs
docs ↗
DS · MLE · C Build a model · E Train at scale
What: Ephemeral managed compute that trains on S3 data, then shuts down.
When: Any custom or built-in training run, billed per second.

Automatic Model Tuning
docs ↗
DS
What: Hyperparameter optimization across many parallel training jobs.
When: Find the best hyperparameters without manual trial and error.

MLflow (was Experiments)
docs ↗
DS · MLE · I Track & govern
Serverless MLflow, 2025
What: Managed MLflow to track runs, metrics, and GenAI traces.
When: Compare runs and record what produced each model.

Training Plans
docs ↗
MLE · Admin · E Train at scale
What: Reserve GPU capacity for a set window, for training jobs or HyperPod.
When: Big runs that can't wait for on-demand capacity.

Debugger & Profiler
docs ↗
DS · MLE · Closed to new
What: Inspect tensors and flag training problems; profile resource bottlenecks.
When: Diagnose vanishing gradients, overfitting, or idle GPUs.

05 Deploy & infer

Real-time Endpoints
docs ↗
MLE · F Serve predictions
What: Always-on HTTPS endpoint with autoscaling.
When: Low-latency predictions with steady traffic.

Serverless Inference
docs ↗
MLE · F Serve predictions
What: Endpoint that scales to zero; no instance type to pick.
When: Spiky or rare traffic that tolerates cold starts.

Asynchronous Inference
docs ↗
MLE · F Serve predictions
What: Queued requests; results written to S3.
When: Large payloads or long processing per request.

Batch Transform
docs ↗
DS · MLE · F Serve predictions
What: Score a whole dataset offline, no persistent endpoint.
When: Nightly or one-off bulk predictions.

Inference Recommender
docs ↗
MLE
What: Load-tests your model across instance types and configs.
When: Pick the cheapest setup that meets latency targets.

Shadow Tests
docs ↗
MLE
What: Mirror live traffic to a new model or config and compare.
When: Validate a change before it serves real users.

Neo
docs ↗
MLE
What: Compiles a trained model for specific cloud or edge hardware.
When: Faster inference on a target chip or device.

06 Monitor & explain

Model Monitor
docs ↗
MLE · Ops · J Model quality · Closed to new
What: Scheduled checks for data drift and quality drift on endpoints.
When: Alert when production data or accuracy shifts.

Clarify
docs ↗
DS · Risk · J Model quality · Closed to new
What: Bias detection and explainability (SHAP), plus foundation model evaluation.
When: Fairness reviews and "why this prediction?" questions.

Augmented AI (A2I)
docs ↗
Ops · G Human review · Closed to new
What: Human review loops on low-confidence predictions.
When: A person must check risky or uncertain outputs.

Model Dashboard
docs ↗
MLE · Risk · J Model quality
What: One view of every model, endpoint, and monitor status.
When: Oversight of all models in production.

07 MLOps & governance

Pipelines
docs ↗
MLE
What: Workflow DAGs: prep, train, evaluate, register, deploy.
When: Automate and repeat the ML workflow end to end.

Model Registry
docs ↗
MLE · I Track & govern
What: Model versions, approval status, and metadata.
When: Gate which version is allowed to deploy.

Projects
docs ↗
MLE · Admin
What: MLOps templates that wire up repos, CI/CD, and pipelines.
When: Standardize how teams ship models.

Lineage Tracking
docs ↗
MLE · Risk · I Track & govern
What: Records links from data to job to model to endpoint.
When: Audits, and reproducing how a model was made.

Model Cards
docs ↗
Risk · I Track & govern
What: Document intended use, risk rating, and evaluation results.
When: Governance and compliance reporting.

Role Manager
docs ↗
Admin · Closed to new
What: Persona-based, least-privilege IAM roles for ML users.
When: Set up DS, MLE, and ops permissions quickly.

RETIRED OR FROZEN: DON'T PLAN NEW WORK ON THESE

Edge Manager
docs ↗
Status: Shut down April 26, 2024.
Instead: AWS IoT Greengrass V2 with ONNX or Neo.

Studio Classic
docs ↗
Status: Past end of maintenance; no new apps can be created.
Instead: the current Studio.

Ground Truth Plus
info ↗
Status: Discontinued June 30, 2026.
Instead: third-party labeling vendors, or Ground Truth if already onboarded.

Training Compiler
docs ↗
Status: No new releases; existing containers get no patches.
Instead: standard training containers and HyperPod recipes.

"Closed to new customers" means accounts that hadn't started using the feature by July 30, 2026 can't onboard; existing users keep access. The same notice covers Amazon Mechanical Turk. Other SageMaker AI features are unaffected.

Sources: SageMaker AI features page (updated Oct 15, 2026) · SageMaker AI Developer Guide